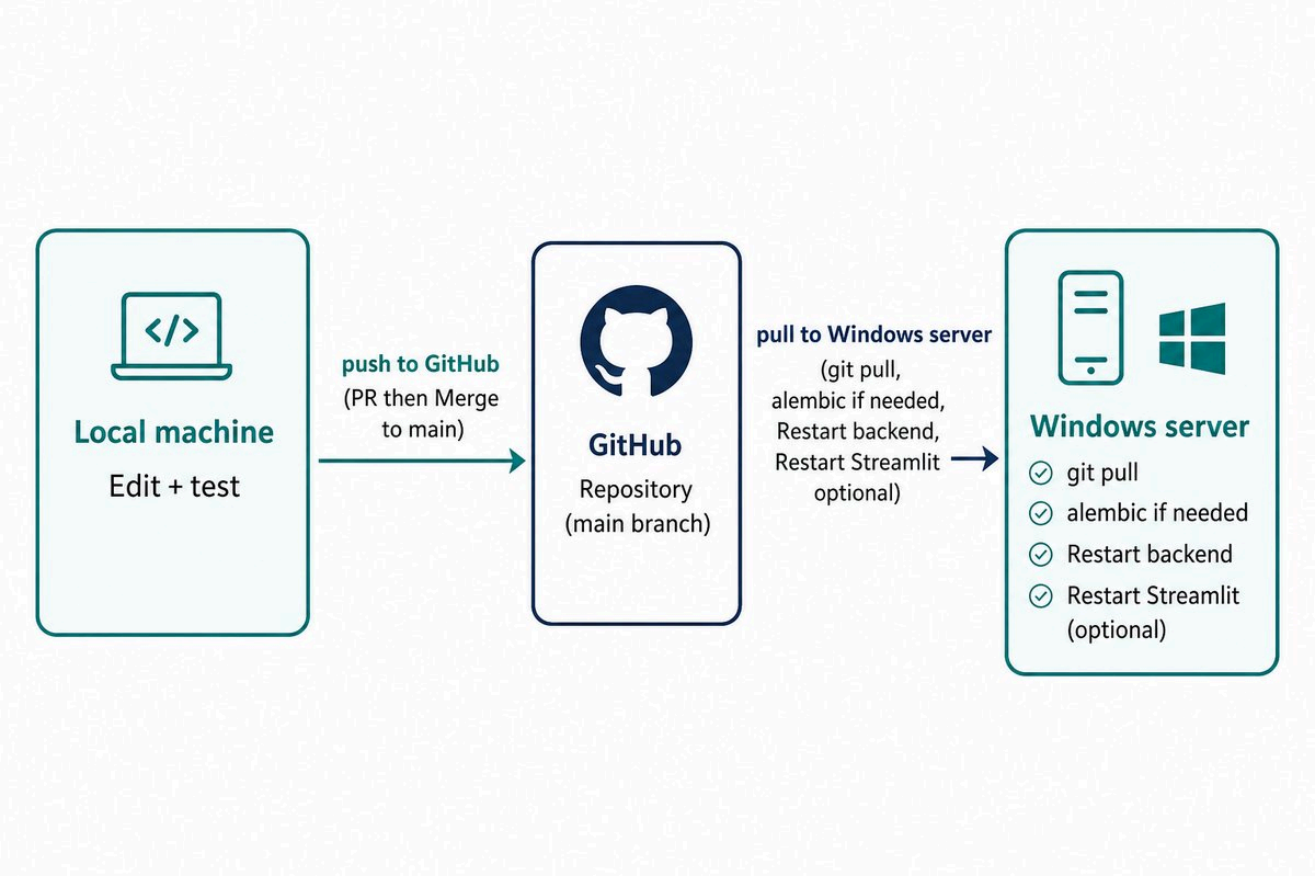
Deploy Guide

> **Audience:** developers / ops
> **Purpose:** ship code from local → GitHub → Windows production server

Deploy path at a glance



Step	Where	What you do
1	Local machine	Edit, test, commit, push branch
2	GitHub	Open PR → `main` → Merge
3	Windows server	`git pull`
4	Windows server	Run Alembic if there are migrations
5	Windows server	Restart `HospitalChatbotBackend`
6	Windows server	Restart Streamlit **only if** dashboard code changed

> **Reliable path:** merge to `main`, then update the server **manually**.
> GitHub Actions auto-deploy may fail (SSH auth) — do not rely on it alone.

1) Local → GitHub

From the project root on your machine:

```
cd E:\4S\hospital_chatbot
git checkout -b <branch-name>    # or switch to an existing branch
# after changes and testing:
git add <files>
git commit -m "short clear message"
git push -u origin <branch-name>
```

Then on GitHub:

Step	Action
1	Open a Pull Request: branch → `main`
2	Review the diff
3	Merge

After merge, production code is on `main`. Continue with the server update below.

2) Server update — backend bot

Connect with AnyDesk / RDP.

Project path: `C:\apps\hospital-chatbot`

```
cd C:\apps\hospital-chatbot
git checkout main
git pull

# Only if requirements.txt changed:
.\venv\Scripts\pip.exe install -r requirements.txt

# Apply any new DB migrations:
.\venv\Scripts\python.exe -m alembic upgrade head

Restart-Service HospitalChatbotBackend
Get-Service HospitalChatbotBackend
Get-Content "C:\apps\hospital-chatbot\logs\stdout.log" -Tail 20 -Encoding UTF8
```

Expected service status: **Running**.

Alembic note (existing databases)

Situation	Command
First time only — DB already has full schema, Alembic never stamped	`alembic stamp head`
Every normal deploy after that	`alembic upgrade head`

Windows services

Service	Purpose
`HospitalChatbotBackend`	Backend / WhatsApp bot
`NgrokTunnel`	Tunnel used for WhatsApp connectivity

> There is **no** Windows service named `HospitalChatbotDashboard`. Do not use that name.

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3) Server update — Streamlit dashboard

Dashboard is **not** a Windows service. Restart it manually after `git pull` when dashboard code changed.

Official URL: `http://196.219.86.38:8501/`

```
cd C:\apps\hospital-chatbot
.\venv\Scripts\streamlit.exe run dashboard/streamlit_app.py --server.address 0.0.0.0 --server.port 8501
```

- Stop any old process on port `8501` first if needed.
- Keep the PowerShell window open while the dashboard should stay online.
- Open the URL and refresh the browser.

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4) Local vs server

Environment	URL	Notes
Local dashboard	`http://localhost:8501`	Usually a different DB → different numbers
Server dashboard	`http://196.219.86.38:8501/`	Official production view
Local API	`http://127.0.0.1:8000`	Dev only

After a production deploy:

- You may stop local `uvicorn` / Streamlit / `cloudflared` with `Ctrl+C`.
- Do **not** stop Streamlit on the server if you want the public dashboard link to keep working.

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5) Local webhook tunnel (optional)

For local WhatsApp webhook testing only:

```
cd E:\4S\hospital_chatbot
.\cloudflared tunnel --url http://127.0.0.1:8000
```

This is for development. Production uses the server setup / tunnel service.

Quick deploy checklist

- [ ] Changes committed and pushed
- [ ] PR merged into `main`
- [ ] Server: `git checkout main` + `git pull`
- [ ] `pip install` only if requirements changed
- [ ] `alembic upgrade head` if migrations exist
- [ ] `Restart-Service HospitalChatbotBackend`
- [ ] Service shows Running; logs look healthy
- [ ] Restart Streamlit only if dashboard files changed
- [ ] Spot-check WhatsApp bot (and dashboard if needed)